

# Deliverable 3

CHU Decision-Support Data Repository

*Results · Storytelling · Dashboard*



| Field       | Details                                     |
|-------------|---------------------------------------------|
| Institution | CESI                                        |
| Group       | Group 2                                     |
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| Instructor  | Boris Ngamamba                              |
| Version     | Deliverable 3 (Final)                       |
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## 1. Executive Summary & Storytelling

### The story we tell CHU decision-makers

*"From fragmented healthcare files to one decision cockpit."*

The CHU group generates data across PostgreSQL patient systems, hospitalization CSV exports, a 25M-row national death registry, facility/professional reference files, and patient satisfaction surveys. Without integration, each department sees only a slice of reality.

Our Big Data decision system answers one question: **Where should CHU invest to improve care access, outcomes, and patient experience?**

#### Chapter 1 — Access to care (Consultations)

We track consultation volumes by hospital, diagnosis, and professional. This reveals which facilities and specialties face the highest demand and where staffing may be insufficient.

#### Chapter 2 — Care intensity (Hospitalizations)

Hospital stays highlight clinical pressure: which diagnoses drive admissions, and which gender/age groups are most affected. Average length of stay signals capacity planning needs.

#### Chapter 3 — Population outcomes (Deaths 2019)

Regional death counts provide a public health baseline. Combined with geo enrichment, CHU can compare regions and prioritize prevention programs.

#### Chapter 4 — Patient experience (Satisfaction)

ESATIS 2019 scores show which regions and hospitals patients recommend least. This complements clinical metrics with perceived quality.

#### Closing recommendation

CHU should maintain the layered DWH (staging → star schema → Power BI), refresh satisfaction annually, and expand geo reference data for complete regional death analytics.

## 2. Methodology

| Phase                                 | Deliverable | Output                                      |
|---------------------------------------|-------------|---------------------------------------------|
| Requirements & conceptual model       | D1          | 4-star schemas, architecture, Talend jobs   |
| Physical implementation & performance | D2          | Hive tables, loads, partitions, benchmarks  |
| Visualization & storytelling          | D3          | Power BI dashboard, narrative, presentation |

### Architecture (recap)

DATA 2024 + PostgreSQL → Talend → Hive Staging → Hive DWH (star schema) → Power BI

## Data sources used (DATA 2024)

| Source                                                        | Role in DWH                      |
|---------------------------------------------------------------|----------------------------------|
| Hospitalisations.csv                                          | FACT_HOSPITALIZATION             |
| etablissement_sante.csv                                       | DIM_HOSPITAL                     |
| professionnel_sante.csv +<br>activite_professionnel_sante.csv | DIM_PROFESSIONAL + facility link |
| deces.csv (~25M rows)                                         | FACT_DEATH                       |
| resultats-esatis48h-mco-open-data-2019.csv                    | FACT_SATISFACTION                |
| PostgreSQL Consultation / Patient / Diagnosis                 | FACT_CONSULTATION, DIM_PATIENT   |

## 3. Requirements for Managing Medical Activities

| ID | Business question                                   | Primary fact table                      |
|----|-----------------------------------------------------|-----------------------------------------|
| B1 | Consultation rate at facility X over period Y       | FACT_CONSULTATION                       |
| B2 | Consultation rate by diagnosis X over period Y      | FACT_CONSULTATION                       |
| B3 | Overall hospitalization rate over period Y          | FACT_HOSPITALIZATION                    |
| B4 | Hospitalizations by diagnosis                       | FACT_HOSPITALIZATION                    |
| B5 | Hospitalizations by gender & age group              | FACT_HOSPITALIZATION +<br>DIM_PATIENT   |
| B6 | Consultation rate per healthcare professional       | FACT_CONSULTATION +<br>DIM_PROFESSIONAL |
| B7 | Deaths by region (2019)                             | FACT_DEATH + DIM_GEO                    |
| B8 | Overall satisfaction by region (2020 target/ESATIS) | FACT_SATISFACTION                       |

## 4. Dashboard Design (Power BI)

### 4.1 Dashboard Pages

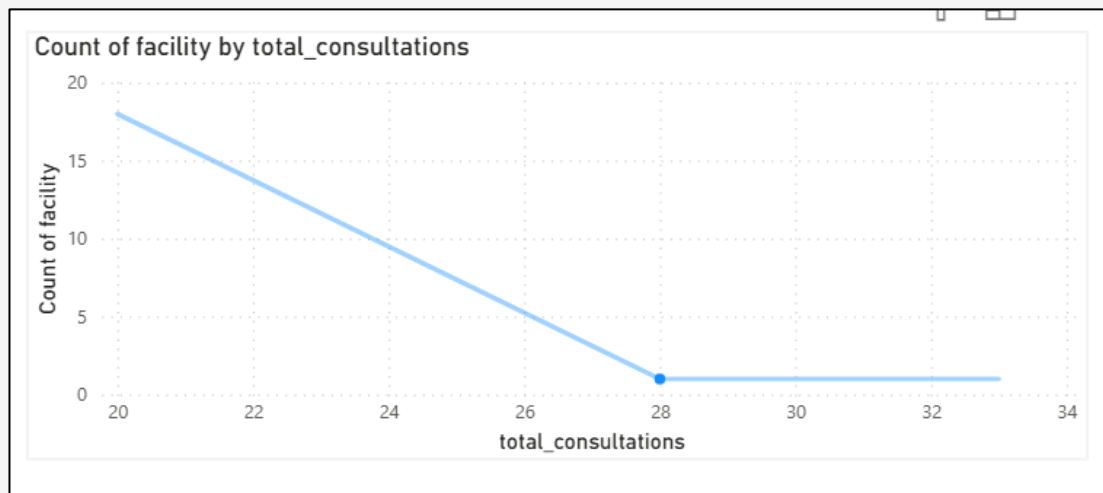
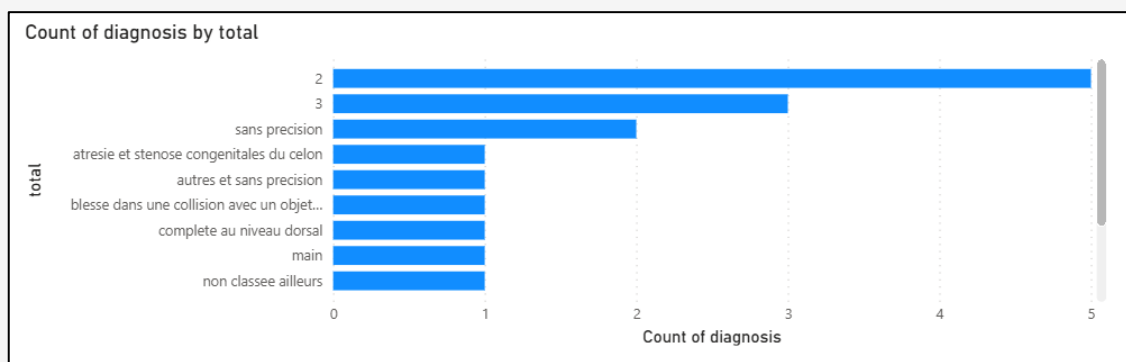
| Page | Title              | Requirements covered |
|------|--------------------|----------------------|
| 1    | Executive Overview | KPI cards + trends   |
| 2    | Consultations      | B1, B2, B6           |
| 3    | Hospitalizations   | B3, B4, B5           |
| 4    | Deaths             | B7                   |
| 5    | Satisfaction       | B8                   |

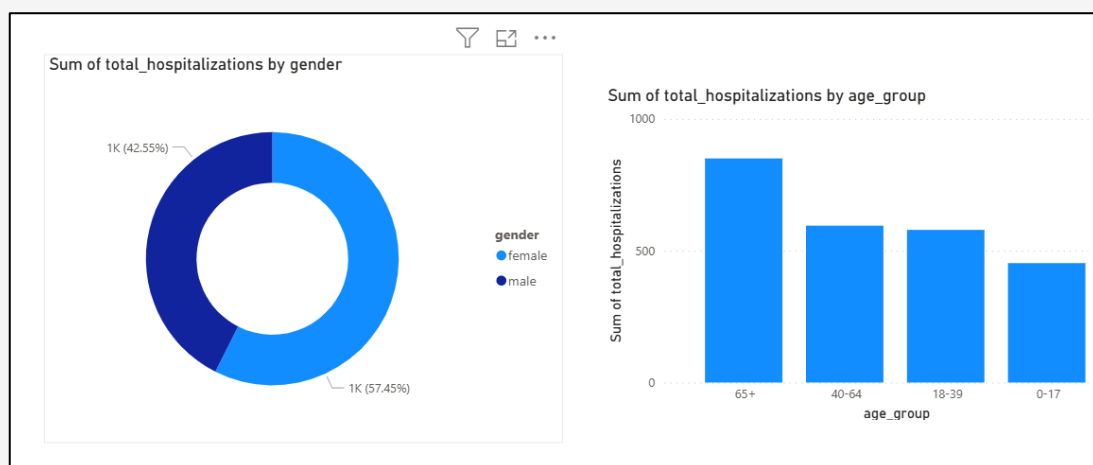
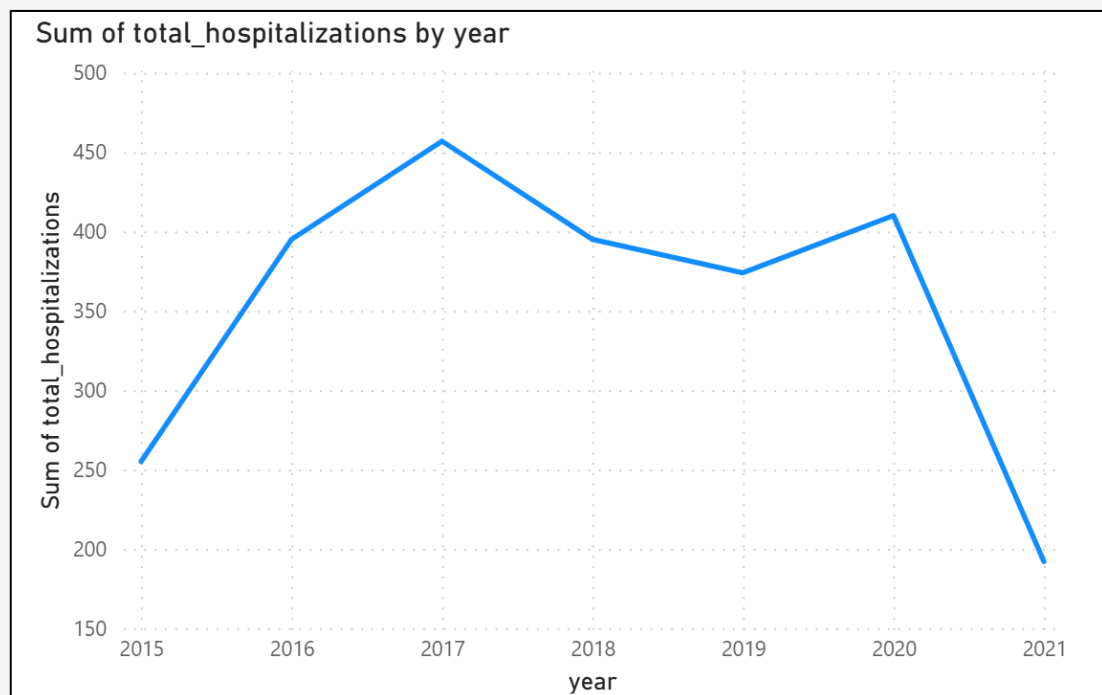
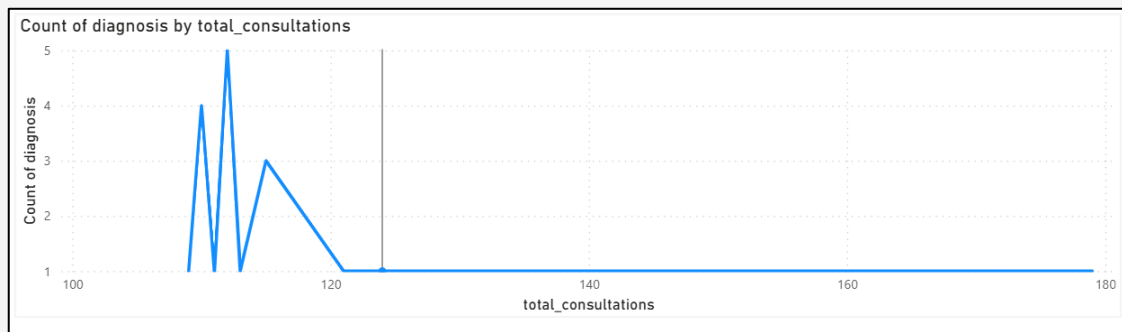
## 4.2 Global Slicers

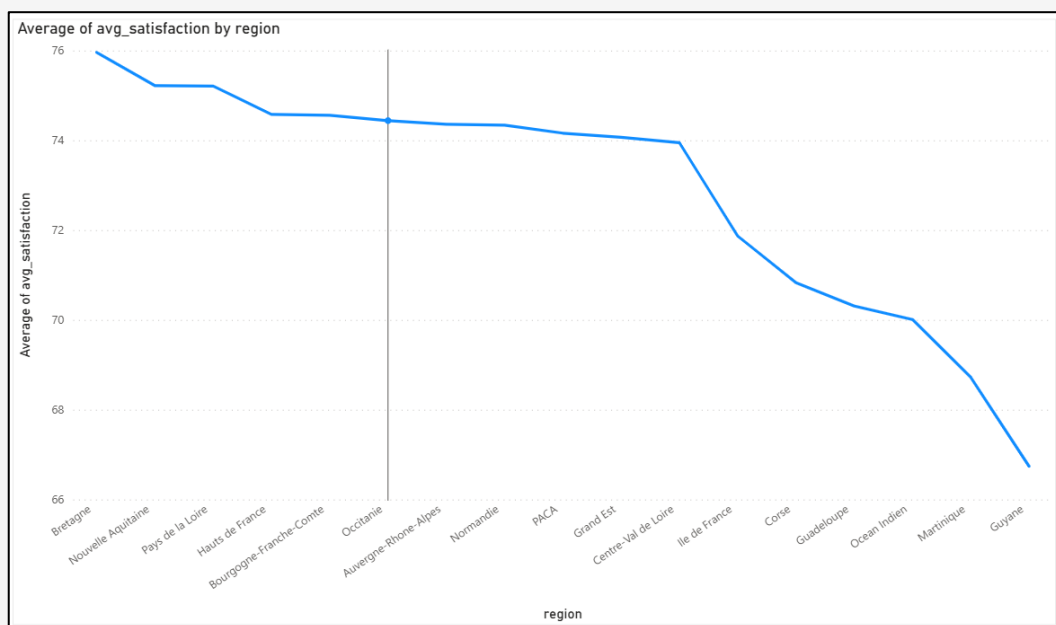
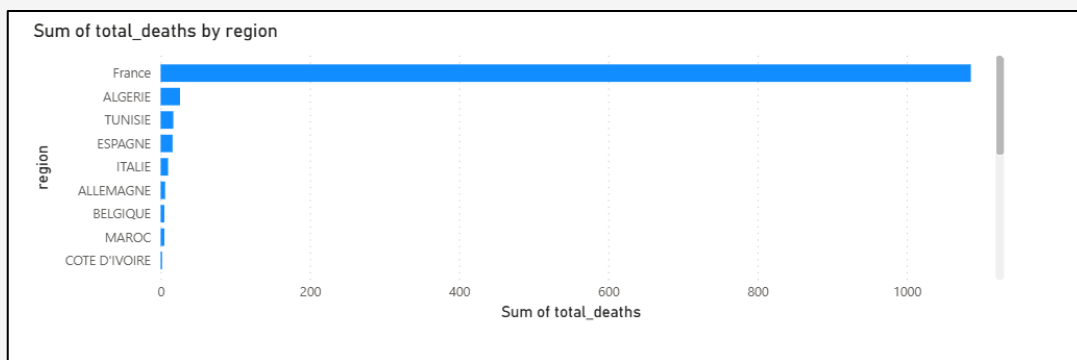
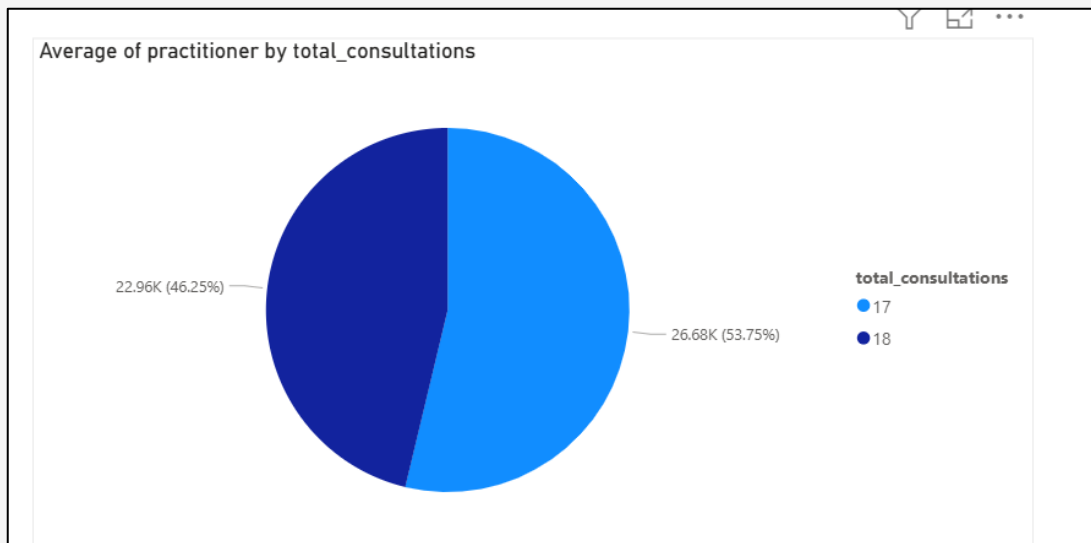
- ▶ Year
- ▶ Region
- ▶ Hospital
- ▶ Diagnosis
- ▶ Gender
- ▶ Age group

## 4.3 Dashboard Screenshots

Insert the Power BI screenshots below after finalizing the dashboard:







## 5. Requirement-by-Requirement: Visual + SQL

### B1 — Consultation rate at facility X over period Y

| Item   | Choice / Detail                                             |
|--------|-------------------------------------------------------------|
| Visual | Line chart (monthly trend) + bar chart (top hospitals)      |
| Why    | Line shows evolution over period Y; bar compares facilities |
| SQL    | power_bi/queries/B1_consultations_by_facility_period.sql    |

### B2 — Consultation rate by diagnosis X over period Y

| Item   | Choice / Detail                                             |
|--------|-------------------------------------------------------------|
| Visual | Matrix or clustered bar (diagnosis × month)                 |
| Why    | Highlights seasonal and clinical patterns for one diagnosis |
| SQL    | power_bi/queries/B2_consultations_by_diagnosis.sql          |

### B3 — Overall hospitalization rate over period Y

| Item   | Choice / Detail                                      |
|--------|------------------------------------------------------|
| Visual | KPI card + column chart by quarter                   |
| Why    | Single KPI for executives; quarters show seasonality |
| SQL    | power_bi/queries/B3_hospitalizations_overall.sql     |

### B4 — Hospitalizations by diagnosis

| Item   | Choice / Detail                                       |
|--------|-------------------------------------------------------|
| Visual | Horizontal bar chart (Top 25 diagnoses)               |
| Why    | Easy ranking of clinical drivers                      |
| SQL    | power_bi/queries/B4_hospitalizations_by_diagnosis.sql |

### B5 — Hospitalizations by gender and age group

| Item   | Choice / Detail                                     |
|--------|-----------------------------------------------------|
| Visual | Stacked bar chart (gender × age_group)              |
| Why    | Shows demographic distribution in one view          |
| SQL    | power_bi/queries/B5_hospitalizations_gender_age.sql |

### B6 — Consultation rate per healthcare professional

| Item   | Choice / Detail                                       |
|--------|-------------------------------------------------------|
| Visual | Bar chart — Top 25 professionals                      |
| Why    | Identifies workload concentration                     |
| SQL    | power_bi/queries/B6_consultations_by_professional.sql |

### B7 — Deaths by region in 2019

| Item   | Choice / Detail                                   |
|--------|---------------------------------------------------|
| Visual | Map (if region geo available) or sorted bar chart |
| Why    | Geographic comparison is the core requirement     |
| SQL    | power_bi/queries/B7_deaths_by_region_2019.sql     |

### B8 — Overall satisfaction by region (2020 target / ESATIS 2019)

| Item   | Choice / Detail                                            |
|--------|------------------------------------------------------------|
| Visual | Bar chart (avg score by region) + table (response counts)  |
| Why    | Score for ranking; counts ensure statistical confidence    |
| SQL    | power_bi/queries/B8_satisfaction_by_region.sql             |
| Note   | ESATIS 2019 campaign used — closest available in DATA 2024 |

## 6. Key Findings & Recommendations

### 6.1 Findings (to validate with your loaded data)

- ▶ **Hospitalizations** (~2,479 rows in DATA 2024) cover multiple years — use year slicers to avoid mixing periods.
- ▶ **Deaths** require the year partition and a complete geo reference for accurate regional totals.
- ▶ **Satisfaction** varies significantly by region; low scores may correlate with lower response volumes.
- ▶ **Consultations** depend on PostgreSQL data and the professional → facility linkage via the activity file.

### 6.2 Recommendations for CHU

| Recommendation                 | Detail                                                                                             |
|--------------------------------|----------------------------------------------------------------------------------------------------|
| Automate monthly Talend jobs   | Schedule staging → DWH refresh to keep dashboards current                                          |
| Complete geo reference         | Replace sample INSEE commune table with full national reference to eliminate UNKNOWN_REGION deaths |
| Row-level security in Power BI | Restrict sensitive patient fields per user role before wider deployment                            |
| Annual satisfaction ingests    | Schedule ESATIS file processing each year once new survey results are published                    |
| Pre-aggregated summary tables  | Build summary tables for frequently used aggregates to reduce Power BI query time                  |



## 7. Technical Demonstration Checklist

| Slot | Duration | Content                                                                    |
|------|----------|----------------------------------------------------------------------------|
| 1    | 2 min    | Context & CHU needs                                                        |
| 2    | 4 min    | Architecture & methodology (D1 recap)                                      |
| 3    | 4 min    | Physical model & performance (D2 — show one before/after benchmark chart)  |
| 4    | 8 min    | Live Power BI walkthrough — all 5 pages, tell the story chapter by chapter |
| 5    | 2 min    | Conclusions & next steps                                                   |

## 8. Appendices

| Artefact                   | Description                                                    |
|----------------------------|----------------------------------------------------------------|
| power_bi/measures_dax.txt  | All DAX measures used in the dashboard                         |
| power_bi/dashboard_spec.md | Full dashboard specification (pages, visuals, slicers)         |
| power_bi/queries/B1–B8.sql | SQL queries behind each business requirement                   |
| Deliverable 1 folder       | Architecture, conceptual model, star schemas, Talend job specs |
| Deliverable 2 folder       | Physical Hive scripts, partitioning, benchmark results         |